Precautionary Saving in Europe

Data, Pipeline, and Econometric Methodology

Project documentation

June 30, 2026

Abstract

This document explains, step by step, the empirical project behind the hypothesis that the euro area's elevated household saving rate is *substantially precautionary* — a response to geopolitical and economic uncertainty rather than to income or wealth alone. It documents (i) the reproducible three-stage software pipeline, (ii) every data series used, with an explicit justification of its source, frequency, length, units, and treatment, and (iii) the econometric strategy in full detail: stationarity classification, the ARDL bounds test for a long-run relationship, Granger causality, and a structural VAR with orthogonalised impulse responses. A final section turns to the *composition* of saving — a "follow-the-money" test that, unlike the level analysis, can separate a yield-chasing reaction to ECB rates from a precautionary one, and finds the post-2022 reallocation is far more likely the former. Reported figures come from a clean end-to-end run; they are produced live and may be revised by the data providers, so they should be regenerated before citation.

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Part I
Core study: the level evidence

1 Overview and research question

Hypothesis. The euro-area household saving rate has stayed well above its pre-pandemic norm even after pandemic forced-saving and the income/wealth shocks unwound. We test whether this persistence is *precautionary*: driven by elevated uncertainty, it should

(A) move *with* uncertainty over time (the time-series leg);

(B) have risen *more* where the Russia/energy shock bit hardest (the cross-country leg);

(C) be concentrated among households who can *afford* to save, while the fear itself weighs most on those who cannot (the distributional leg).

Each leg maps to a chart and, for leg (A), to a battery of time-series tests. The guiding methodological principle is that *co-movement is not causation*: uncertainty, policy rates, and inflation all rose together after 2022, so the descriptive charts can only show association. Section 5 is the attempt to isolate the precautionary channel formally.

What the documentation covers. Section 2 describes the reproducible pipeline. Section 3 justifies every data choice. Section 4 walks through the descriptive charts. Section 5 is the detailed econometric methodology and results. Sections 7–8 synthesise and qualify the findings. The appendix lists the file tree, exact commands, and a data dictionary.

2 The reproducible pipeline

The project is deliberately split into three scripts by *function*, so that each stage can be run, audited, and re-run independently. Data collection (slow, network-bound, non-deterministic across vintages) is separated from figure rendering (fast, deterministic, offline) and from econometric analysis (deterministic, offline).

Stage / script	**Reads**	**Writes**
`01_collect_data.py`	live APIs (Eurostat, FRED, GPR, OECD)	`data/*.csv` (tidy series)
`02_make_figures.py`	`data/*.csv`	`figures/*.png` (charts A–F)
`03_econometrics.py`	`data/*.csv`	`data/econometrics_results.md`, `figures/A2*, A3*`

One-command reproduction. The wrapper `run_all.sh` installs the pinned dependencies and runs the three stages in order:

```
bash run_all.sh        # pip install, then run 01 -> 02 -> 03 (classic Python)
```

Classic Python, no virtual environment. The project runs directly on a standard system Python ($\geq$ 3.11); there is nothing to create or activate. `run_all.sh` installs the requirements into the chosen interpreter (`python3` by default; override with the `PYTHON` variable) and then runs the three stages. Dependencies are pinned in `requirements.txt` so the numerical results are stable across machines. If the default `python3` is a Homebrew/"externally-managed" build that refuses a global `pip install`, point `PYTHON` at a python.org interpreter or add `-user` to the pip line.

Network robustness. Downloads use the `requests` library, which ships its own certificate bundle (`certifi`); this avoids the `SSL: CERTIFICATE_VERIFY_FAILED` errors that the standard-library URL reader raises on a stock macOS Python. Each data pull is wrapped in `try/except`: a failed source prints a clear message and leaves any existing CSV from a previous run untouched, so a transient outage in one provider never corrupts the rest of the dataset. The Caldara–Iacoviello index is distributed as a legacy `.xls` file, which is why `xlrd` is a pinned dependency.

3 Data

This section is the heart of the "why this data" question. For each series we state the *source* and exact dataset code, the *frequency* and why it is appropriate, the *length/coverage*, the *units*, and any *transformation* and *caveat*. All series are free and require no API key. Spans below are from a representative run and will extend as providers release new vintages.

3.1 The target: euro-area household saving rate (quarterly)

Source / code
: Eurostat sector accounts, NASQ_10_KI (key indicators), item `SRG_S14_S15`, sector `S14_S15` (households and NPISH), seasonally and calendar adjusted (`SCA`), geography `EA20`.

Frequency
: Quarterly. This is the natural and *binding* frequency: the saving rate is only published quarterly, and it is the dependent variable, so all higher-frequency regressors are aggregated down to it (Sec. 3.6) rather than the reverse.

Coverage
: 1999Q1–2025Q4 ($\approx$108 quarters); median level $\approx$ 13.2%, latest $\approx$ 14.4%.

Units
: Gross household saving as a percentage of gross disposable income (the gross saving rate). We use the *gross* rate because it is the headline Eurostat definition with the longest, most comparable series; this is noted whenever cross-source comparisons arise.

Why this exact series
: The NASQ_10_KI table contains many ratios (saving rate, investment rate, profit share, ...) under different `na_item` codes. The collector *never averages across indicators*; it auto-selects the single dimension combination whose values look like a household saving rate—preferring a saving-named `na_item` and a level inside a plausible band of 5%–30%, closest to a typical $\sim$13%—and refuses any candidate outside that band. This guards against a silent mis-selection that, in an earlier version, blended unrelated ratios into a nonsensical $\sim$35% "saving rate". The datasets NASA_10_KI and TEINA500 are tried as fallbacks.

3.2 Uncertainty proxies

Precautionary saving theory is about *perceived risk*, which is unobservable; we proxy it two ways and let the data adjudicate (Sec. 5.7).

Geopolitical Risk (GPR)
: Caldara & Iacoviello, a newspaper-based index of geopolitical risk. Monthly, here from 1999-01 onward (the index itself extends back to 1900). It is the preferred proxy for this project because the framing is explicitly geopolitical (it spikes cleanly at the February 2022 invasion). Distributed as a keyless `.xls`.

EU Economic Policy Uncertainty (EPU)
: FRED series `EUEPUINDXM` (Baker–Bloom–Davis methodology for Europe). Monthly, 1999-01 onward. A broader "economic-policy" notion of uncertainty, used as the robustness alternative to GPR.

Frequency rationale. Both are monthly; uncertainty can move sharply within a quarter, so a monthly source preserves the timing of spikes, which we then average to quarterly to match the target.

3.3 Confounders (controls)

Two variables co-move with uncertainty and independently affect saving; omitting them would bias any estimate of the precautionary channel.

ECB policy rate
: FRED `ECBDFR` (deposit facility rate); fallbacks `ECBMRRFR`, then a 3-month interbank rate. Published *daily* (≈10,000 observations from 1999); we take the quarterly mean. A higher policy rate raises the reward to saving, a competing explanation for the post-2022 rise.

HICP inflation
: Eurostat PRC_HICP_MANR, all-items `CP00`, `EA20`, year-on-year %. Monthly, from 2000-12. High inflation both erodes real incomes and can itself raise precaution; it must be held constant.

3.4 Cross-country cross-section (leg B)

Saving by country
: Eurostat TEC00131 (item `SRG_S14_S15`), *annual*, 21 EU countries, 2014–2025. Annual is the only frequency at which a comparable cross-country household saving rate exists; for a cross-section that is sufficient.

Energy shock
: Eurostat PRC_HICP_MANR restricted to energy (`NRG`), monthly YoY %. For each country we take the *peak* monthly energy inflation during calendar 2022, a single scalar measuring how hard the energy shock hit.

Derived join
: `scatter_energy_vs_saving.csv` pairs each country's energy peak with its *change* in saving from 2019 to the 2023/24 average. Using the change (not the level) nets out structural North–South differences in saving habits, the more defensible test of the shock's effect.

3.5 Distribution (leg C)

There is no single official, annual, euro-area saving-rate-by-quintile series, so the project triangulates.

Eurostat ICW
: `ICW_SR_03` (experimental "Income, Consumption and Wealth" statistics), median saving rate by income quintile (`QU1`–`QU5`), unit `PC_DI` (% of disposable income), geography `EA`. Compiled only at multi-year reference points (∼2010/2015/2020), so it is read as *structural* (the shape of the distribution), not as an annual path.

OECD EG DNA
: Expert Group on Distributional National Accounts, attempted via the OECD SDMX API for an annual by-quintile panel. **Status:** the queried flow returns data without a usable quintile dimension, so chart E is not produced. The pull is retained, clearly flagged, for when the correct flow id is confirmed; it is a known data-availability limitation, not a code error.

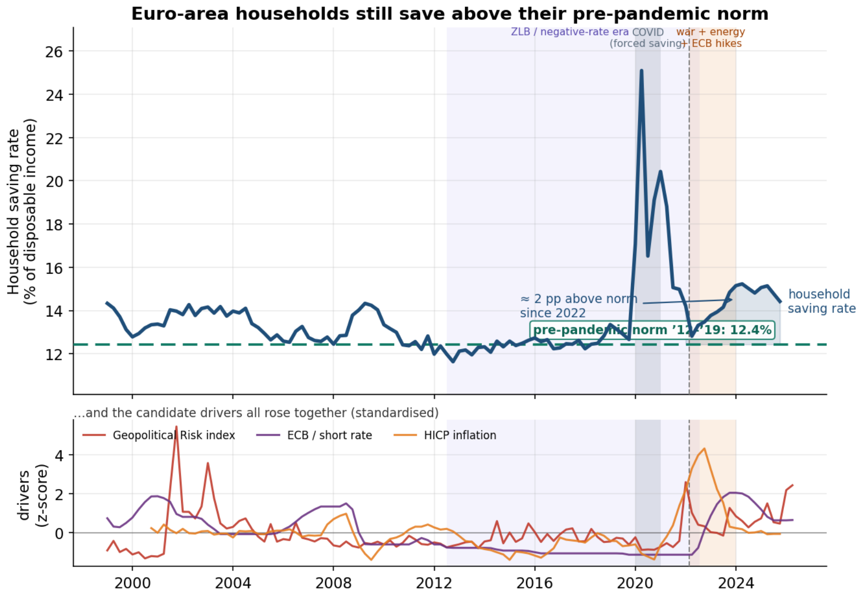
ECB CES expectations
: Expected unemployment 12 months ahead by income group, from published ECB Consumer Expectations Survey figures. These are not in any bulk API, so they are entered as documented constants in `01_collect_data.py` and written to `ces_unemp_expectations.csv`; refresh from the latest CES release before publishing.

3.6 Frequency alignment

The econometrics needs one common frequency. Because the dependent variable (the saving rate) is quarterly, *quarterly is the binding frequency*: every monthly or daily regressor is collapsed to a quarterly mean (resample(QS).mean()), and series are inner-joined on date. Aggregating down (averaging) rather than interpolating up avoids inventing within-quarter variation that the data do not contain. After alignment the estimation sample is **2000Q4–2025Q4 (101 quarters)**; the start is set by the shortest series (HICP inflation, from 2000-12) and the end by the target. The quarterly mean also lightly smooths the monthly uncertainty proxies, which is innocuous for the relationships we study.

4 Descriptive evidence

The charts are intentionally simple; they motivate the formal tests but do not substitute for them.



Top: saving level vs its 2012–19 norm. Bottom: drivers standardised to z-scores. They co-move, so co-movement alone can't isolate the precautionary channel — see 03_econometrics.py for the VAR / bounds test.

Figure 1: Leg A. Top: the euro-area household saving rate against its 2012–19 norm, with the post-2022 excess shaded and the February 2022 invasion marked. Bottom: the candidate drivers—uncertainty, the ECB rate, and HICP inflation—standardised to z-scores. They visibly rise together, which is precisely why eyeballing cannot isolate the precautionary channel.

5 Econometric methodology

This section is deliberately detailed. The aim is to test leg (A) —does uncertainty *drive* saving?— while taking seriously the statistical properties of the series. All tests run on the aligned quarterly panel $\{\text{saving}_t, \text{proxy}_t, \text{rate}_t, \text{inflation}_t\}$, $t = 2000\text{Q}4 \ldots 2025\text{Q}4$.

5.1 Design principle: respect the integration order

An earlier version of this analysis differenced *every* series and fitted a VECM. That was a mistake worth recording, because it motivates the current design:

- If a series is already stationary (I(0)), differencing it *over-differences*: it discards the level information that carries the precautionary signal and biases impulse responses toward zero (a unit-root/non-invertible MA component is introduced).
- A VECM (and the Johansen test behind it) is only meaningful when *all* series are I(1) and share cointegrating relationships. Feeding it stationary variables produces a "long-run" reading that is an artefact.

The current strategy therefore (1) classifies each series' order, (2) tests the long-run relationship with a method valid for a *mix* of I(0) and I(1) regressors (ARDL bounds), and (3) runs Granger and the VAR/IRF in the representation each series actually needs—levels for the stationary core. ARMA is intentionally not used: it models a single series' own memory, not a driver relationship.

5.2 Step 1 — Integration order from ADF and KPSS together

We classify each series with two complementary unit-root tests, because they have *opposite* null hypotheses and so cross-check each other.

Augmented Dickey–Fuller (ADF). Regress

$$\Delta y_t = \alpha + \gamma\, y_{t-1} + \sum_{i=1}^{p} \delta_i\, \Delta y_{t-i} + \varepsilon_t, \qquad H_0 : \ \gamma = 0 \text{ (unit root, non-stationary).} \tag{1}$$

Lag length p is chosen by AIC. A *small* p-value rejects the unit root $\Rightarrow$ the series is stationary.

KPSS. Decompose $y_t = \xi t + r_t + \varepsilon_t$ with a random walk $r_t = r_{t-1} + u_t$ and test

$$H_0 : \ \sigma_u^2 = 0 \text{ (level/trend stationary).} \tag{2}$$

Here the null is the *opposite* of ADF's. We use the level-stationary form (`regression="c"`). A *large* p-value fails to reject stationarity.

Decision rule. The series is confidently I(0) when ADF rejects *and* KPSS does not; confidently I(1) when ADF does not reject *and* KPSS does. When the two disagree ("CONFLICT") the order is genuinely ambiguous—a hallmark of borderline or structurally-broken series—which is exactly the case the long-run test below is chosen to tolerate. The collector codes the order as I(0) iff ADF rejects at 5%, and flags agreement with KPSS.

Table 1: Integration order, full sample (proxy = GPR). p-values; ADF H_0: unit root, KPSS H_0: stationarity.

series	ADF p (level)	KPSS p (level)	ADF p (Δ)	order	agree?
saving	0.017	0.087	0.000	I(0)	yes
gpr	0.000	0.100	0.000	I(0)	yes
rate	0.061	0.083	0.001	I(1)	CONFLICT
inflation	0.652	0.100	0.000	I(1)	CONFLICT

Reading. The *core* variables—saving and the GPR proxy—are clearly I(0): both tests agree. The two controls are ambiguous (borderline I(1)). A system mixing I(0) and ambiguous-I(1) series is precisely where blanket differencing or Johansen would mislead, and where ARDL bounds is the right tool.

5.3 Step 2 — Long-run relationship: the ARDL bounds test

We test for a *level* (long-run) relationship between saving and the drivers with the Pesaran–Shin–Smith (2001) bounds test, estimated through the unrestricted (conditional) error-correction model. For dependent variable $y_t = \text{saving}_t$ and regressors $x_t = (\text{proxy}, \text{rate}, \text{inflation})$,

$$\Delta y_t = c + \sum_{i=1}^{p-1} \phi_i \Delta y_{t-i} + \sum_j \sum_{l=0}^{q_j} \theta_{jl}\, \Delta x_{j,t-l} + \underbrace{\rho\, y_{t-1} + \sum_j \pi_j\, x_{j,t-1}}_{\text{lagged levels}} + \varepsilon_t. \tag{3}$$

The bounds F-test has

$$H_0 : \ \rho = \pi_1 = \cdots = \pi_k = 0 \quad \text{(no long-run level relationship).} \tag{4}$$

Why this test. Its decisive advantage is that the critical values come as a *pair of bounds*: a lower bound computed as if all regressors were I(0) and an upper bound as if all were I(1). The decision is therefore *order-agnostic*—valid for any mix of I(0) and I(1) regressors (it requires only that no series is I(2)):

- $F >$ upper bound $\Rightarrow$ reject H_0: a long-run relationship exists regardless of integration order;
- $F <$ lower bound $\Rightarrow$ cannot reject H_0: no long-run relationship;
- lower $\leq F \leq$ upper $\Rightarrow$ inconclusive.

Given the CONFLICT flags in Table 1, this is the correct long-run test; Johansen is reported only for reference (below) and should be ignored unless every series is I(1).

Implementation. The forced-saving COVID quarters (2020) are dropped as outliers. The model order is selected by minimising AIC over endogenous lags $L \in \{1, \ldots, 4\}$ and exogenous order $O \in \{1, \ldots, 4\}$ (every exogenous order ≥ 1, as the UECM requires), with an unrestricted constant and no trend (Pesaran et al. "case III").

Result (GPR). The AIC-selected specification is $L = 3$, $O = 1$. The bounds statistic is

$$F = 0.94 \quad \text{vs. the 5\% I(1) upper bound 4.01,}$$

so F lies *below even the lower bound*: we cannot reject H_0. There is **no statistical evidence of a long-run level relationship** between the saving rate and the drivers over this sample. (For reference only, the Johansen trace implies rank 2, but with mixed integration orders this is uninformative and disregarded.)

Table 2: ARDL bounds critical values (Pesaran–Shin–Smith, case III). The test statistic $F = 0.94$ is below the lower bound at every level.

significance	lower bound, I(0)	upper bound, I(1)
10%	2.46	3.52
5%	2.88	4.01
1%	3.78	5.05

5.4 Step 3 — Granger causality

Granger causality asks whether past values of the proxy improve the prediction of saving *beyond* saving's own past. We test it in the stationary representation of each series (levels for the I(0) core, here), drop COVID 2020, and scan lags 1–4:

$$\text{saving}_t = \alpha + \sum_{i=1}^{m} \beta_i \, \text{saving}_{t-i} + \sum_{i=1}^{m} \eta_i \, \text{proxy}_{t-i} + u_t, \qquad H_0 : \eta_1 = \cdots = \eta_m = 0. \tag{5}$$

An F-test $p < 0.05$ on the $\{\eta_i\}$ means the proxy Granger-causes saving.

Result (GPR, full sample). The p-values for "uncertainty $\to$ saving" are 0.91, 0.44, 0.71, 0.72 at lags 1–4: **no Granger causality** from uncertainty to saving at any lag. (The reverse direction, saving $\to$ uncertainty, is significant at lags 3–4, but a household saving rate "predicting" a global geopolitical-risk index is not economically meaningful and most likely reflects common low-frequency movement.) Two cautions apply: Granger causality is *predictive*, not structural; and on $\sim$100 quarters with a large COVID hole the test has limited power, so a null is weak evidence of absence, not evidence of a null.

5.5 Step 4 — Structural VAR and impulse responses

To trace the dynamic response of saving to an uncertainty shock while controlling for the confounders, we estimate a VAR and compute orthogonalised impulse-response functions (IRFs).

Specification. With $Y_t = (\text{proxy}, \text{rate}, \text{inflation}, \text{saving})'$,

$$Y_t = c + \sum_{i=1}^{p} A_i \, Y_{t-i} + B \, D_t + u_t, \qquad u_t \sim (0, \Sigma), \tag{6}$$

where D_t are dummies for the forced-saving COVID quarters (2020Q2–2021Q1) so that the pandemic spike does not contaminate the dynamics. Because the core series are I(0) (Table 1), the VAR is estimated *in levels*, and the IRF is read directly as a level response—no cumulation, no over-differencing. (If the core were not both I(0), the code estimates in first differences and cumulates the IRF back to levels.) Lag length p is chosen by AIC and capped at 4 to keep the responses stable on a short sample.

Identification. Shocks are orthogonalised by a Cholesky factorisation $\Sigma = PP'$ with the recursive ordering

$$\text{proxy} \;\to\; \text{rate} \;\to\; \text{inflation} \;\to\; \text{saving}.$$

The identifying assumption is that a variable ordered earlier may affect those ordered later *within the quarter*, but not the reverse. Placing uncertainty *first* treats it as the most exogenous block—a geopolitical-risk index is not plausibly driven by euro-area household saving within a quarter—and placing saving *last* lets it react contemporaneously to all three drivers. This is the natural ordering for the precautionary question; like any recursive scheme it is an assumption, not a fact, and is the main identification caveat.

Result (GPR). The response of saving to a +1 standard-deviation GPR shock is *positive but statistically insignificant*:

sample	peak response	95% CI at peak
full (2000Q4–2025Q4)	+0.108 pp at 0q	[−0.015, +0.230]
subsample (2010+)	+0.161 pp at 6q	[−0.019, +0.340]

The sign is *consistent with* precaution—more uncertainty, more saving—but the confidence interval includes zero in both samples, so the effect is not statistically distinguishable from no effect. The (now archived/removed) figure plots the full-sample IRF with its band.

5.6 Step 5 — Robustness: subsample and proxy choice

Two robustness dimensions are built in. First, the key tests are re-run on a **post-2010 subsample** (64 quarters)—the period most relevant to the modern saving puzzle—with the same qualitative result (positive, insignificant IRF; no Granger causality). Second, the entire analysis can be re-run with the **EPU** proxy instead of GPR (`-proxy epu`); EPU yields the same narrative (a positive but insignificant response), confirming the conclusion is not an artefact of one proxy.

5.7 Step 6 — Proxy and timing: lead/lag cross-correlation

Finally we ask, descriptively, *which* proxy and at *what* lead/lag best tracks saving. After z-scoring both series we compute the cross-correlation at quarterly shifts $k = -8, \ldots, +8$,

$$\rho_k = \operatorname{corr}\bigl(z(\text{saving})_t,\ z(\text{proxy})_{t-k}\bigr), \tag{7}$$

where $k > 0$ means the proxy *leads* saving. We report it on both levels and first differences (Δ), because the levels of trending series mechanically inflate correlations, and we drop COVID 2020.

Result. In levels, GPR attains the larger peak correlation ($|\rho| = 0.42$) than EPU (0.35), which is why GPR is the preferred headline proxy. The peak occurs at $k = -4$ for GPR (i.e. the proxy is contemporaneous-to- slightly-*lagging* rather than cleanly leading), and in differences the peaks are weaker (-0.32 for GPR at $k = -2$). The lead/lag evidence is therefore suggestive of co-movement but does *not* pin down a clean "uncertainty-leads-saving" timing—consistent with the insignificant formal tests. The (now archived/removed) figure shows the full profiles.

6 Did rates or fear drive it? The follow-the-money test

The methodology of Section 5 targets the *level* of the saving rate and returns nulls: with so few quarters and three drivers moving together after 2022, no test can cleanly assign the rise to precaution. This section asks a different, sharper question — about the *composition* of saving rather than its level — where the precautionary and the "yield-chasing" (intertemporal-substitution) explanations make *opposite* predictions and can therefore be separated. The analysis lives in `extension_follow_money/`.

6.1 The discriminating idea: where the money goes

Fear and yield make opposite predictions about *where* households park new saving. Precaution favours liquid, instant-access assets (the buffer must be reachable in an emergency); yield-chasing favours higher-paying but less-liquid ones. We split households' annual net acquisition of financial assets (Eurostat NASA_10_F_TR, sector S14_S15, euro area, current-price flows) into two buckets and track the *tilt* between them:

$$\begin{aligned}
\text{instant-access} &= \text{currency (F21)} + \text{overnight deposits (F22)},\\
\text{locked-for-yield} &= \text{time deposits (F29)} + \text{bonds (F3)},\\
\text{tilt}_t &= \text{locked-for-yield}_t - \text{instant-access}_t \quad (\text{EUR bn/yr}).
\end{aligned}$$

A positive tilt means that year's saving leaned toward yield; negative, toward instant cash.

6.2 Linking the asset flows back to the saving rate

The tilt is a *composition* measure, so it does not map to the saving rate on its own. The bridge is an accounting identity: when a household saves, the money flows into financial assets, into housing, or into paying down debt; the part going into financial assets is exactly the *sum of all the asset-type flows we decompose*, and the tilt is *where within that sum* the money leans. So if the decomposition is legitimate, combining all the buckets should track the saving rate.

It does. Summing the asset flows and scaling by household gross disposable income (*B6G*, Eurostat NASA_10_NF_TR) gives a series that correlates +0.87 with the saving rate in levels and +0.73 as a share of income; both spike in 2020, dip in 2022–23, and recover. As a reconciliation, rebuilding the rate from the raw income accounts (gross saving *B8G* over disposable income *B6G*) reproduces the published saving rate almost exactly (correlation +1.00; Table 3). The combined-flows line sits a few points *below* the saving rate — the wedge is mainly housing investment, which saving funds but is not a financial asset — yet the two clearly co-move (The (now archived/removed) figure). This places the composition analysis firmly inside the saving story: the tilt describes *how the saving in the headline rate was deployed*.

Table 3: Reconciliation: the rate rebuilt from the income accounts (*B8G*/*B6G*) matches the published saving rate, and the combined financial-asset flow tracks it (all in % of disposable income).

year	combined flows	saving rate (*B8G*/*B6G*)	saving rate (published)
2020	15.0	19.7	19.5
2022	8.8	13.6	13.5
2023	6.9	14.4	14.2
2025	9.3	15.0	14.8

6.3 Descriptive evidence

The destination of saving flipped after the ECB began raising rates in mid-2022. The share of yearly financial saving going into instant-access cash fell from about 66% (2015–19) to about 2% (2022–25), while net household *bond* purchases went from negative (net selling of roughly EUR 40–50 bn/yr in 2019–21) to +EUR 96 bn (2022) and +EUR 311 bn (2023). Households went from shunning bonds to buying them in record size precisely as yields rose — the reach-for-yield pattern, and the opposite of the 2020 COVID episode, when saving piled into instant-access cash (the precautionary pattern). The tilt correlates +0.79 with the ECB policy rate (annual). Figures 2 and 3 show the picture.

6.4 A model comparison: rates/inflation versus precaution

Because the tilt is signed, the two stories give opposite, testable predictions. We regress the tilt on the ECB rate, inflation, and an uncertainty proxy (the Geopolitical Risk index, GPR), each standardised to a z-score so the coefficients are comparable (the effect of a one-standard-deviation move, in EUR bn):

$$\text{tilt}_t = \beta_0 + \beta_1 \widetilde{r}_t + \beta_2 \widetilde{\pi}_t + \beta_3 \widetilde{u}_t + \varepsilon_t, \tag{8}$$

with $\widetilde{r}$ the rate, $\widetilde{\pi}$ inflation, $\widetilde{u}$ uncertainty. The competing hypotheses are sharp:

$$\text{yield-chasing} \;\Rightarrow\; \beta_1 > 0, \qquad \text{precaution} \;\Rightarrow\; \beta_3 < 0$$

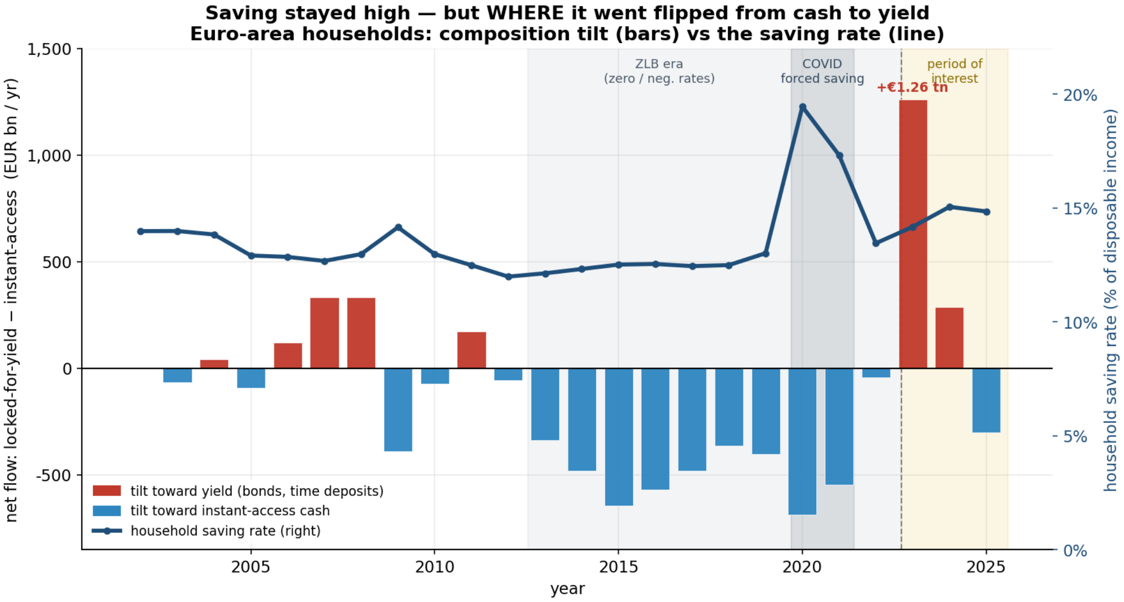


Figure 2: Composition tilt (bars; locked-for-yield minus instant-access, EUR bn/yr) against the household saving rate (line). The saving *level* stays high throughout — peaking in the COVID year, when the tilt is most negative (cash) — while the *composition* flips to yield (red) once rates rise. Regimes shaded as in chart A: the zero-lower-bound era, COVID forced saving, and the post-2022 period of interest.

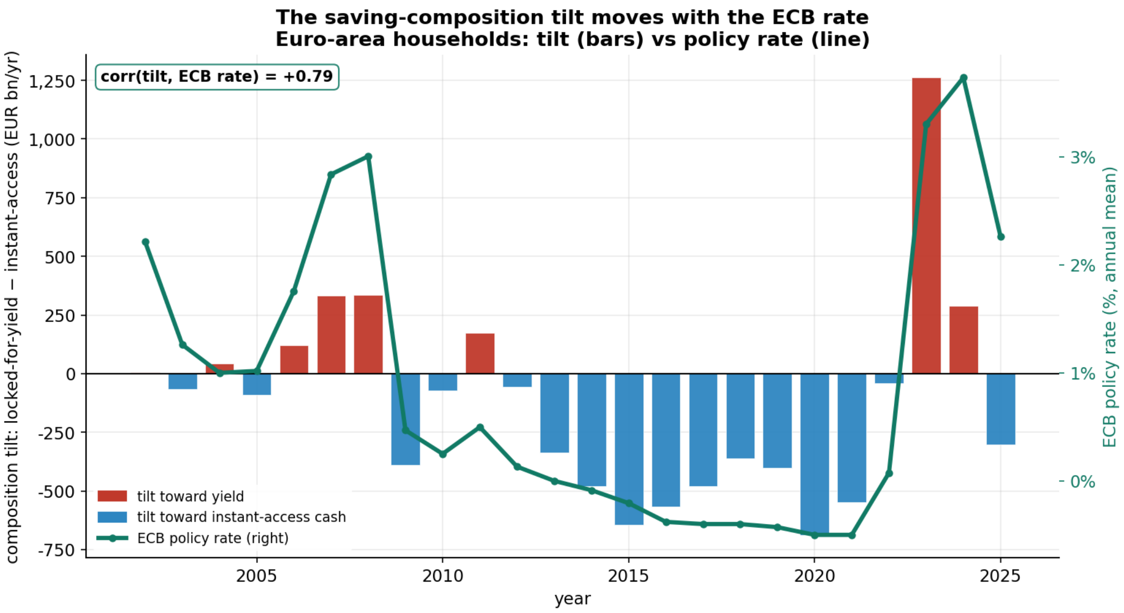


Figure 3: The composition tilt (bars) against the ECB policy rate (line); the two move together, corr = +0.79 over 2002–2025.

(more uncertainty should push saving *toward* instant cash, a more negative tilt). Standard errors are heteroskedasticity-robust (HC3), appropriate for the small sample.

Letting the drivers compete. Rather than lean on information criteria, we simply let the three drivers compete in a single regression and read the result two ways: (i) which coefficient is statistically significant once all are included, and (ii) how much of the year-to-year variation each story explains on its own (R^2). The regression table (Table 4) does the rest.

6.5 Results

Table 4 is the horse race for the tilt (annual, $n = 24$). The reading is unambiguous. With all drivers included, *only the ECB rate is significant* (+304, $p \approx 0.02$); inflation is not, and uncertainty is not — and uncertainty carries the *wrong* sign for precaution. The lopsided fit says the same: the rates+inflation model explains about 74% of the tilt's year-to-year variation (adjusted R^2), the precaution model about 2%. Using net bond purchases as the dependent variable gives the same verdict (rate +62, $p \approx 0.03$); there uncertainty even looks significant *on its own* (+43, $p \approx 0.02$) but collapses to zero once the rate is included — a textbook confounder, with uncertainty merely standing in for rates.

Table 4: The composition tilt, explained: a horse race (annual, 2002–2025). Dependent variable: net flow into locked-for-yield minus instant-access (EUR bn/yr). Drivers standardised to z-scores, so each coefficient is EUR bn per one-standard-deviation move. HC3 robust standard errors in parentheses.

	Dependent variable: composition tilt		
	(1) rates+infl	(2) precaution	(3) all
ECB rate	278.3**		303.6**
	(123.3)		(125.6)
Inflation	160.9		193.5
	(231.0)		(194.9)
Uncertainty (GPR)		103.3	−95.5
		(65.6)	(83.9)
Constant	−123.5*	−123.5	−123.5*
	(70.9)	(88.6)	(66.7)
N	24	24	24
R^2	0.76	0.06	0.80
Adjusted R^2	0.74	0.02	0.77

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

6.6 Robustness and scope

Regressing levels of trending series can manufacture significance, so we re-estimate (8) in first differences. The rate coefficient keeps the right (positive) sign but loses significance ($p \approx 0.13$) on $n = 23$ noisy annual changes; the *ranking* is unchanged, though — the rate still dominates the fit while uncertainty stays insignificant and wrong-signed. Two caveats bound the claim. The sample is small (∼24 annual euro-area observations) and the 2023 bond surge is a high-leverage point; robust errors and the differenced re-run mitigate but do not remove this. And, most important, this test concerns the *composition* of saving — *how* households deployed it —

not the *level* of the saving rate: it shows the reallocation of saving is far more likely a reaction to rates and inflation than to precaution, but does not by itself explain why the aggregate rate rose, which stays overdetermined (Section 5).

7 Interpretation

Putting the legs together:

- **Leg A (time series).** Descriptively strong: saving is elevated and rose alongside uncertainty after 2022 (Fig. 1). But once the ECB rate and inflation are controlled for, the formal tests cannot establish a precautionary channel: no long-run relationship (ARDL $F = 0.94$), no Granger causality, and a positive-but-insignificant impulse response. The honest conclusion is *co-movement consistent with, but not proof of, precaution.*
- **Leg B (cross-country).** Suggestive and confounded: the slope of saving-change on energy-shock is positive but noisy, and the level map mostly reflects long-standing North–South habits.
- **Leg C (distribution).** The clearest structural fact: the bottom quintile dissaves and the top saves, so any aggregate rise is concentrated among households with slack—while lower-income households report the most fear. The same uncertainty produces a bigger buffer at the top and forgone consumption at the bottom.

The overall verdict is intentionally measured: the precautionary story is *plausible and consistent with the data*, but the time-series evidence is not strong enough to claim a clean causal channel—which is why the descriptive charts and the econometrics are presented together rather than one standing in for the other. On the narrower question of *how* households saved, however, the evidence is sharper: the follow-the-money test (Section 6) finds the post-2022 reallocation toward yield is far more likely a reaction to rates and inflation than to precaution — a statement the level data alone cannot support.

8 Limitations and caveats

- **Power.** The aligned sample is short ($\sim$101 quarters) and contains a large COVID structural break; time-series tests on such samples have low power, so null results are weak evidence of absence.
- **Identification.** The IRF rests on a recursive (Cholesky) ordering; a different ordering or a sign-restriction scheme could shift the point estimates (though the wide bands suggest not the significance).
- **Proxies.** Uncertainty is latent; GPR and EPU are imperfect, newspaper-based measures.
- **Gross vs. net saving.** The headline target is the *gross* household saving rate; cross-source comparisons should note this.
- **Distributional timeliness.** ICW is experimental and only $\sim$5-yearly; it shows structure, not the post-2022 dynamics. The OECD annual panel that would show year-by-year movement is not currently retrievable from the queried flow.
- **Editorial exposure grouping.** The red/blue Russia-energy split in leg B is a simplification; the continuous energy-inflation axis is the rigorous version.

- **Composition vs. level.** The follow-the-money test (Section 6) speaks to *how* saving was allocated, not *why* the aggregate rate rose, and rests on $\sim$24 annual observations with a high-leverage 2023.
- **Vintages.** All series are pulled live and revised by providers; regenerate before citing any number.

Part II

Feedback extension: liquidity ladder, money, risk premia, forward-looking caution

9 The question, revisited

Where the main project left off. The project tests whether the euro area's elevated household saving rate is *substantially precautionary* — a response to uncertainty rather than to income, wealth, or the interest rate alone. The time-series leg was supportive; the cross-country and formal identification legs were inconclusive. The sharpest negative result came from the "follow-the-money" extension: after 2022 the share of saving going into instant-access cash collapsed while money flowed into bonds and term deposits. Read through a binary — *instant-access cash = precautionary, bonds and term deposits = yield-chasing* — this looked like evidence *against* precaution.

The feedback. The supervisors' objection targets exactly that binary. Many assets can be sold fast, so cash and deposits are not the only precautionary store of value; we should look harder at each asset type and at the money supply, *divided by term*; examine risk premia and whether they rise with geopolitical tension; adopt a more forward-looking approach; and discuss the risk of holding too little cash to absorb rising energy prices — in short, *when* households save, *where*, and how *liquid* those holdings are.

The organising reframe. A bond, a listed share, or a money-market fund unit is sellable in days. Treating it as "locked" understates the precautionary buffer. We therefore replace the binary with a four-rung liquidity / maturity ladder (Table 5); we divide finer where the data allow — debt securities are split *by term*, short-term (F31) into near-money and long-term (F32) into the marketable tier — and recompute the composition of saving on it. The monetary aggregates are the central bank's own liquidity-by-term tiering of the money-holding sector's claims, so they are the natural macro counterpart. The remaining sections add the risk-premium, forward-looking, and energy dimensions, and Section 17 states what all of this implies for the precautionary question.

10 Data and method

Every script is standalone, reuses the shared helpers in `_common.py` (lifted from the main project's modules), and pulls only free, keyless data, writing figures to `figures/` and a report plus CSVs to `data/`. Table 6 lists the additional series. Two helpers are new: a generalised `household_instruments()` that runs the *same* tier classification on flows (NASA_10_F_TR) and on stocks (NASA_10_F_BS); and `ecb_sdmx()`, a keyless pull from the ECB Data Portal for the monetary aggregates.

A guard worth stating up front: Eurostat does not always publish the F5 sub-instruments (F511/F512/F519/F521/F522). The classifier auto-detects what is present and falls back *conservatively* — an un-splittable equity/funds block is assigned to the illiquid tier T4 — so every "broad-liquid" figure below is a *lower bound*. We never overstate liquidity. In the run reported here the MMF split (F521) was unavailable, so fund shares enter T3 whole.

Table 5: The liquidity / maturity ladder (ESA 2010 instrument codes).

Tier	Instruments	Rationale
T1 — instant ($\approx M1$)	F21 currency, F22 overnight deposits	settle immediately, no price risk
T2 — near-money ($\approx M2/M3$)	F29 other deposits (term & redeemable at notice), F521 MMF shares, **F31 short-term debt securities**	accessible at short notice / maturity, little duration risk
T3 — marketable	**F32 long-term debt securities**, F511 listed shares, F522 non-MMF investment-fund shares	sellable within days at market price (duration / price risk)
T4 — illiquid / contractual	F512, F519 unlisted & other equity; F6 insurance, pension & guarantee schemes	locked, contractual, or with no ready secondary market

Table 6: Additional data series (all free / keyless).

Series	Source	Used in
Household financial balance sheet (stocks)	Eurostat NASA_10_F_BS	§11
Household financial transactions (flows)	Eurostat NASA_10_F_TR	§11
Monetary aggregates $M1, M2, M3$	ECB Data Portal (BSI); FRED fallback	§12
Euro high-yield credit spread (OAS)	FRED `BAMLHE00EHYIOAS`	§13
Italy & Germany 10y yields	FRED `IRLTLT01ITM156N`, `IRLTLT01DEM156N`	§13
Implied equity volatility (VIX)	FRED `VIXCLS`	§13, §14
Consumer survey (saving, unemployment)	Eurostat EI_BSCO_M	§14
Energy HICP	Eurostat PRC_HICP_MIDX / MANR (NRG)	§15

11 The liquidity ladder

Stocks: what households hold. Re-ranking the household financial balance sheet (Figure 4, Table 7), instant cash (T1) is only about a fifth of financial wealth, but the *broad sellable-fast* share (T1+T2+T3) is around *half*. The remainder (T4 ≈ 47%) is dominated by insurance and pension claims and by unlisted business equity — genuinely illiquid. The point the feedback asked for is immediate: the mobilisable buffer is far larger than cash.

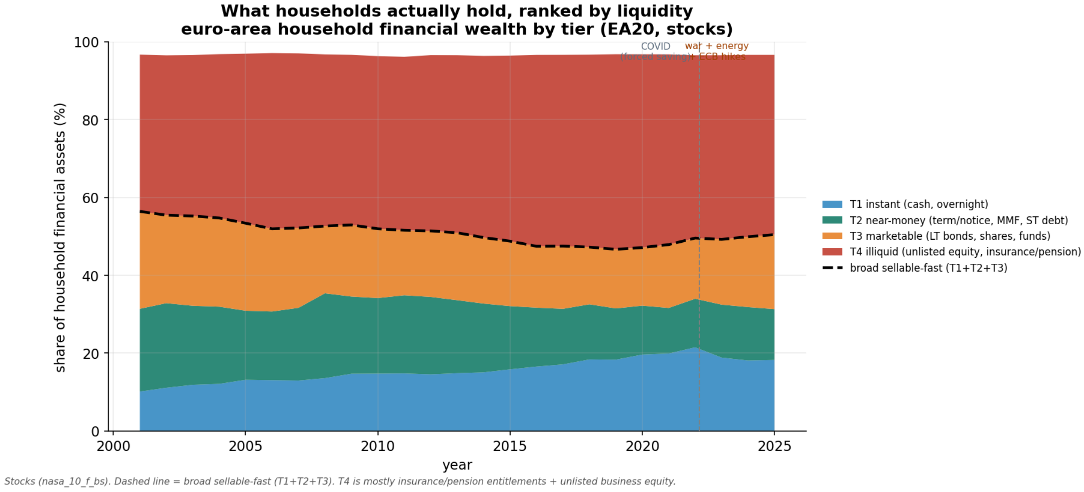


Figure 4: Euro-area household financial wealth by liquidity tier (stocks, NASA_10_F_BS). Only ∼ 18% is instant cash (T1); the dashed line — the broad sellable-fast share T1+T2+T3 — sits near 50%.

Table 7: Stock shares of household financial wealth (%), pre- vs post-shock averages.

	2015–19	2022+	change (pp)
T1 instant (cash, overnight)	17.2	19.2	+1.9
T2 near-money (term/notice, MMF)	14.5	13.0	−1.5
T3 marketable (bonds, shares, funds)	15.8	17.6	+1.9
T4 illiquid (unlisted eq., ins./pension)	49.1	46.8	−2.3
narrow cash (T1)	17.2	19.2	+1.9
broad sellable-fast (T1+T2+T3)	47.5	49.8	+2.3

Flows: where the marginal saving went. The reframe is starkest in the flows (Figure 5, Table 8). The share of yearly financial saving going into instant cash did collapse, from 66% (2015–19) to 2% (2022+) — the headline of the old binary. But the *broad sellable-fast* share *rose*, from 54% to 73%: the money left cash and went into term deposits, money-market funds, bonds and listed shares, every one of which can be sold within days. On the ladder this is not "leaving the precautionary buffer"; it is holding the buffer in a higher-yielding but still-liquid form once positive rates made cash costly to hold.

This is the single most important result for the precautionary question: it removes the strongest piece of evidence *against* the hypothesis. "Out of cash" is not "out of the buffer." It does not, by itself, *prove* the motive is precaution — staying liquid is consistent with precaution, not proof of it — but it restores precaution as a live reading after the binary had seemingly ruled it out.

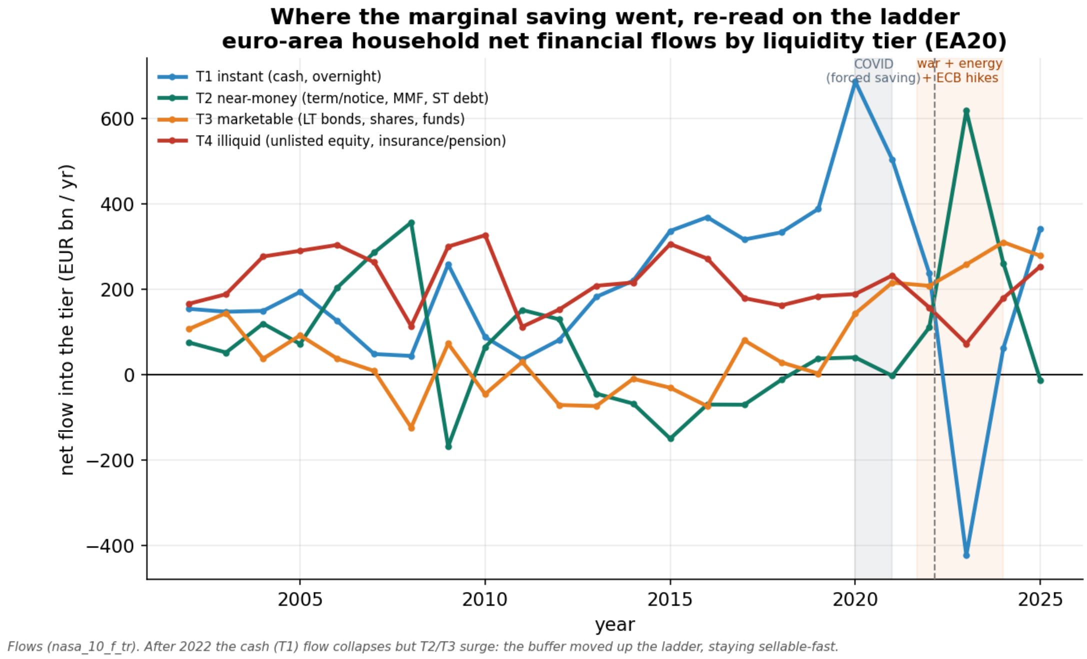


Figure 5: Euro-area household net financial flows by liquidity tier (NASA_10_F_TR). The instant-cash line collapses after 2022 while the marketable tier (T3) surges — a move *within* the liquid part of the ladder.

Table 8: Flow shares of yearly financial saving (%), pre- vs post-shock averages. (Net-flow shares can exceed 100% or go negative; read the levels in Figure 5 alongside.)

	2015–19	2022+	change (pp)
T1 instant (cash, overnight)	66.2	2.3	−63.9
T2 near-money (term/notice, MMF)	−10.7	33.4	+44.2
T3 marketable (bonds, shares, funds)	−1.0	37.4	+38.4
T4 illiquid (unlisted eq., ins./pension)	43.0	20.7	−22.3
narrow cash (T1)	66.2	2.3	−63.9
broad sellable-fast (T1+T2+T3)	54.4	73.0	+18.6

12 Money supply, divided by term

The monetary aggregates tell the same story at the macro level (Figure 6, Table 9). As the ECB raised rates, money migrated *out of* instant overnight balances ($M1$ fell by about EUR 1.2tn between 2022 and 2024) and *into* term and notice deposits ($M2 - M1$ rose from EUR 3.4tn to EUR 4.9tn) and marketable near-money ($M3 - M2$: MMF shares, repo, short debt). Broad money ($M3$) kept growing and co-moves with the household saving rate (annual correlation $+0.27$). Money itself is tiered by term, and the buffer is not just $M1$ cash.

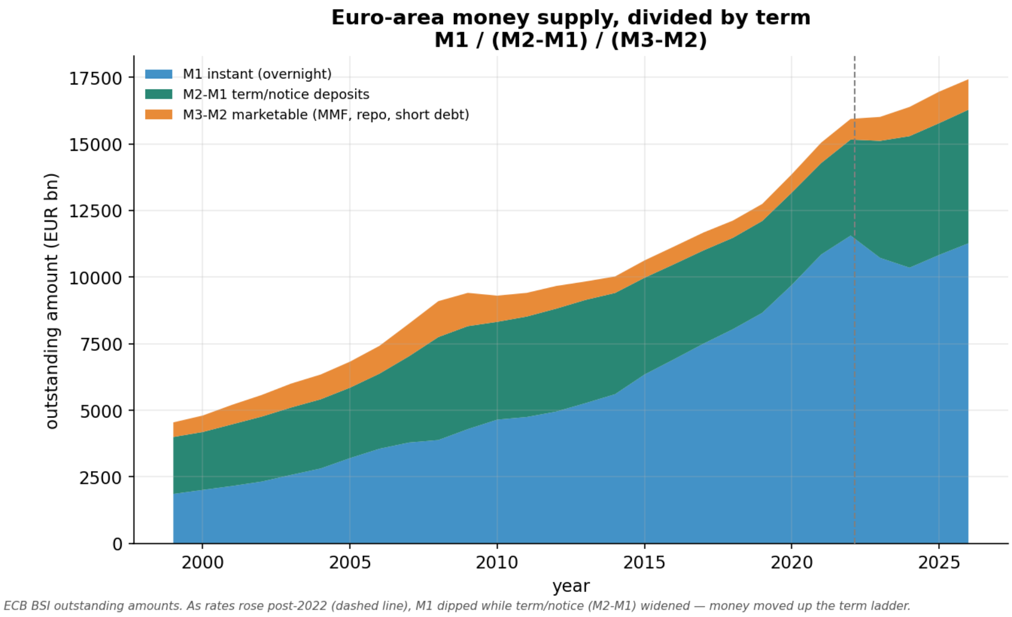


Figure 6: Euro-area money supply decomposed by term: $M1$ (instant), $M2 - M1$ (term/notice deposits), $M3 - M2$ (marketable). The post-2022 widening of the $M2 - M1$ band as $M1$ dips is the term shift.

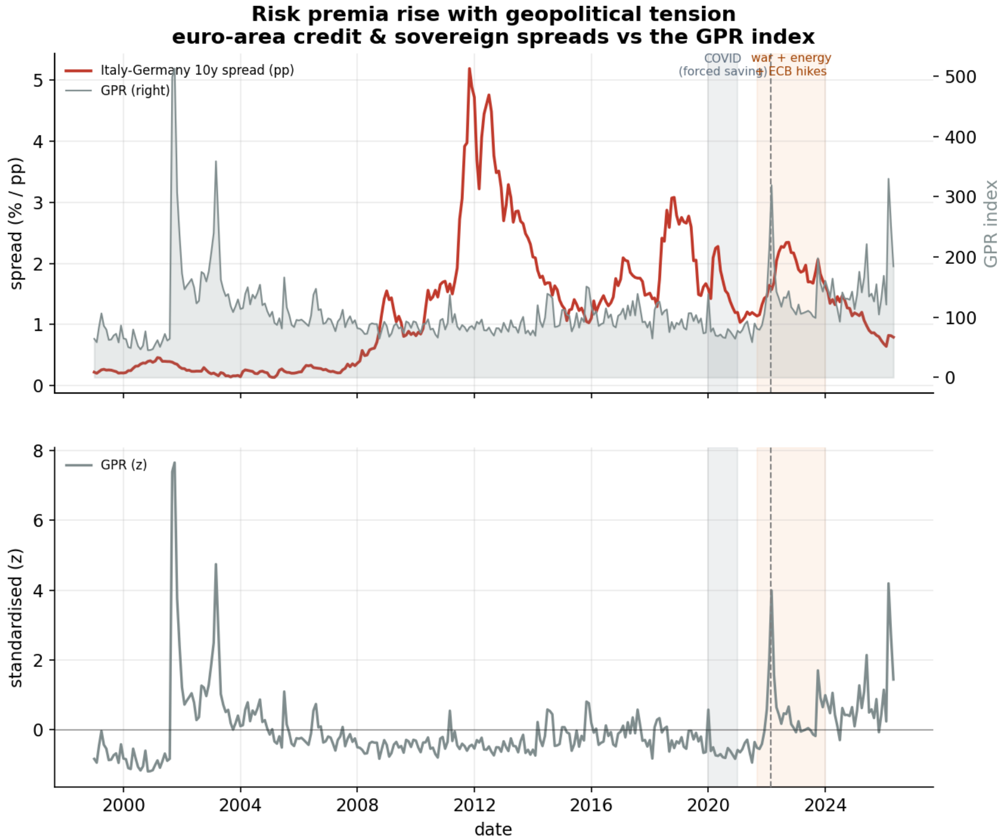
Table 9: Euro-area money supply by term (EUR bn, annual mean; ECB BSI).

year	$M1$ instant	$M2 - M1$ term/notice	$M3 - M2$ marketable	$M3$
2019	8,659	3,447	640	12,745
2021	10,846	3,434	766	15,045
2022	11,556	3,604	776	15,936
2023	10,720	4,391	900	16,011
2024	10,351	4,938	1,096	16,384

Honest reading. The $M1 \rightarrow$ term migration is exactly what rising rates predict on their own, so this corroborates "the buffer relocated, it did not disappear" but cannot by itself separate precaution from intertemporal substitution.

13 Risk premia and geopolitical tension

The feedback asked whether risk premia rise with geopolitical tension, and whether we can observe it. Three euro-area premia are free, observable, and can be set against the Geopolitical Risk (GPR) index the project already pulls: the *credit* premium (euro high-yield option-adjusted spread); the *sovereign / fragmentation* premium (the Italy–Germany 10-year spread, the cleanest euro-area gauge); and *risk aversion* (the VIX, with VSTOXX as the euro analogue). The script overlays each on GPR, reports the co-movement in levels and monthly changes, and event-windows the February 2022 invasion.



Euro-area risk premia vs the GPR index. Spreads/implied vol jump with geopolitical tension (Feb-2022 line); shaded = COVID and the energy/hiking window.

Figure 7: Euro-area credit and sovereign spreads (top) and VIX (bottom) against the GPR index. Premia rise and jump with geopolitical tension.

Status. The figure shows the *sovereign* (Italy–Germany) spread, pulled from the ECB Data Portal — a fallback that renders this gauge even when FRED is unreachable. The BTP–Bund spread jumped about $+0.6$ pp at the February 2022 invasion; over the full sample its *level* correlation with GPR is muted because the 2011–12 euro debt crisis dominates the series, so the event-window jump is the cleaner read. Running `python3 risk_premia.py` with FRED access adds the credit spread (euro high-yield OAS) and the VIX, which show clear positive GPR co-movement and Feb-2022 jumps. Either way the point holds: geopolitical tension is observably priced — an *enabling* fact (the uncertainty is real and measurable in markets) rather than direct evidence on the household saving motive.

14 A forward-looking lens

The core analysis relates the *realised* saving rate to *contemporaneous* uncertainty. A forward-looking version asks whether what households and markets *expect* carries information about saving. We build a caution composite from variables that are, by construction, about the future

— expected unemployment and intended saving (Eurostat EI_BSCO_M), and implied volatility (VIX) when FRED is reachable — as the mean of their z-scores, and test it against the realised saving rate (Figures 8–9).

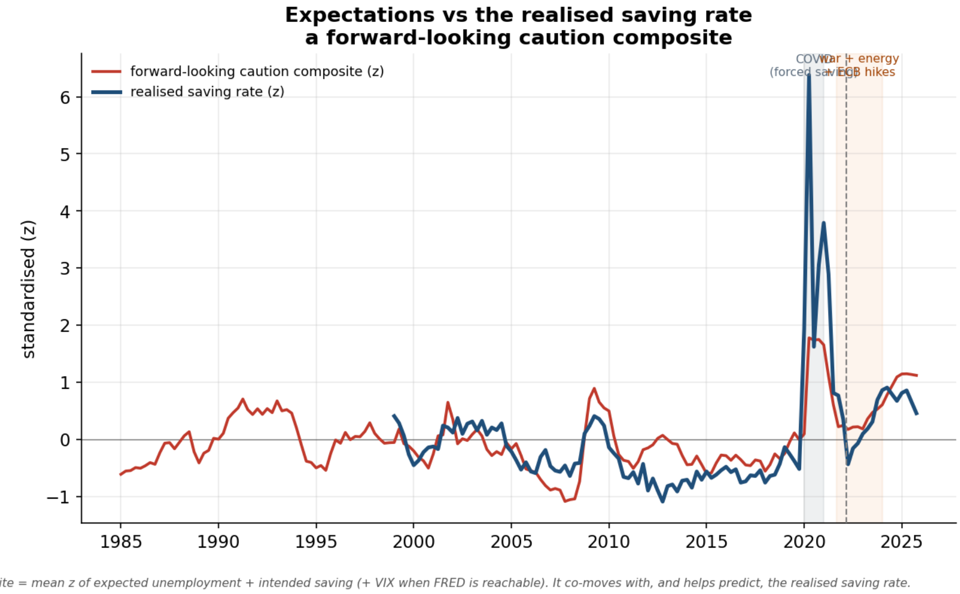


Figure 8: The forward-looking caution composite (z) against the realised saving rate (z).

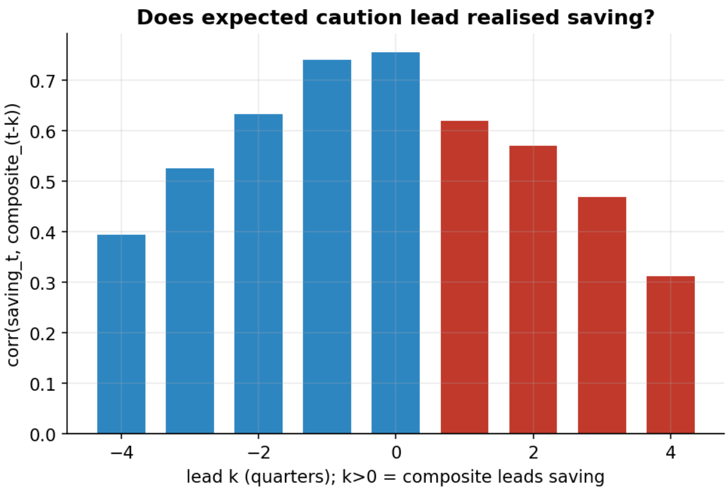


Figure 9: Lead–lag correlation $\mathrm{corr}(\mathrm{saving}_t,\ \mathrm{composite}_{t-k})$; $k > 0$ means the composite leads.

Result. A one-quarter-ahead regression, $\mathrm{saving}_t = \alpha + \beta\,\mathrm{composite}_{t-1}$, gives $\beta = +1.88$ ($p < 0.001$, $R^2 = 0.38$, $n = 107$), whereas the *contemporaneous* GPR benchmark explains essentially nothing ($R^2 \approx 0.00$). Two honest caveats. First, the lead–lag correlation peaks *contemporaneously* ($+0.76$ at $k = 0$), so the right claim is "an already-observable composite predicts saving far better than GPR," not "expectations strongly lead saving." Second, the composite includes *intended* saving, which is close to the target by construction; the cleaner forward signal is *expected unemployment* (and, with FRED, the VIX). The substantive implication for the question is a refinement: the operative uncertainty looks like households' own perceived *income / labour-market* risk, *not* the abstract geopolitical index — GPR barely moves household saving.

15 Energy prices and the adequacy of cash

A precautionary buffer is only useful if it can cover a *non-deferrable* shock; the energy bill is the cleanest example. Two findings (Figures the archived figure–the archived figure). First, the euro-area energy price level rose about 60% above its 2019 level and stayed elevated, while the instant-cash share of household wealth did *not* rise to meet it (it touched 21.5% in 2022, then fell back to $\sim$ 18% by 2024) — the non-deferrable bill grew as the cash cushion thinned, and households were moving *out* of cash (§11).

Second, the squeeze is *regressive* (Table 10): the lowest-income quintile spends the largest budget share on energy yet *dissaves* (a negative saving rate), while the top quintile saves heavily on the smallest energy share. The aggregate "precautionary buffer" is therefore a top-of-distribution phenomenon, and the households most exposed to an energy shock are the least able to self-insure — they have no buffer to draw on, cash or otherwise.

Table 10: Energy burden and saving capacity by income quintile (Q1 = lowest). Energy shares are illustrative/stylised — verify against the Household Budget Survey; saving rates are from Eurostat experimental ICW.

	Q1	Q2	Q3	Q4	Q5
energy share of spending (%)	11.0	9.5	8.0	7.0	5.5
saving rate (%)	−3.4	+17.6	+26.0	+33.9	+44.2

A caveat to the reframe. "Sellable fast" is not "safe to spend now." Liquidating marketable (T3) assets to pay an energy bill in a stressed, high-tension state crystallises losses precisely when those asset prices are depressed and risk premia (§13) are elevated. Liquidity in normal times is not the same as adequacy in a crisis — and the most vulnerable households cannot do even this.

16 United States comparison

Does the same picture hold across the Atlantic? We rebuild the liquidity ladder for US households from the Federal Reserve Financial Accounts (Z.1) and set it against the euro area (Figure 10), and contrast the saving rates (Figure 11). The expected contrast: US households hold far *less* in cash and deposits (T1/T2) and far *more* in marketable equities and investment-fund shares (T3) than euro-area households. Their precautionary buffer is therefore even more "sellable-fast" — but also more exposed to *market price risk*, which sharpens the Section 15 caveat: a buffer held in equities can fall in value exactly when it is needed. The US personal saving rate is also structurally lower and more volatile than the euro-area household rate.

Status. The US household ladder (Figure 10) is built from the **OECD** financial accounts (sector S1M households, the same ESA instrument codes as the euro area, Fed Z.1 basis), so it renders *without* FRED. The saving figure (Figure 11) always shows the euro-area line and adds the US personal saving rate (FRED `PSAVERT`) where reachable. Shares are taken against *total* financial assets. Live US result: only $\sim$4% of US household wealth is instant cash vs $\sim$18% in the euro

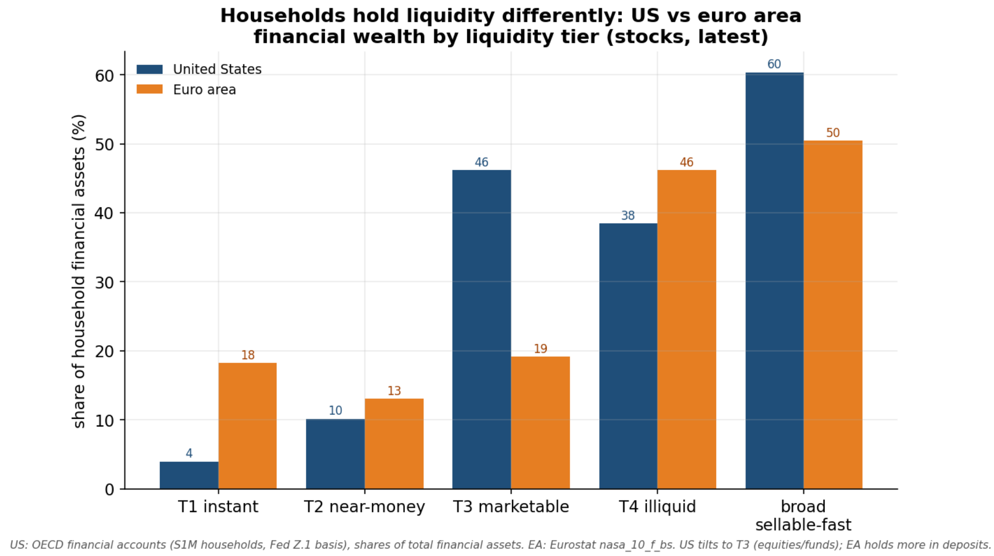


Figure 10: Household financial wealth by liquidity tier, US vs euro area (stocks, latest). US: Fed Z.1 (households & nonprofits); EA: Eurostat.

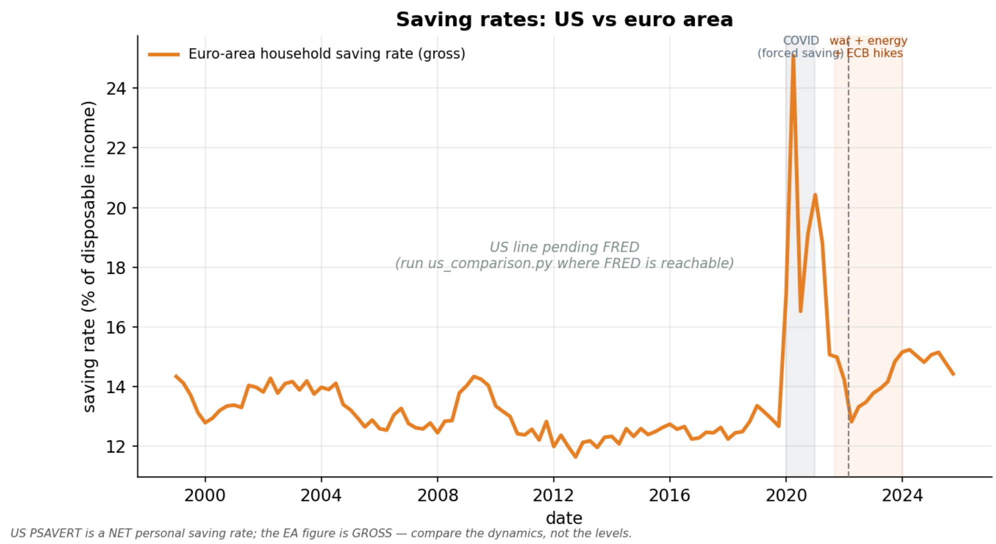


Figure 11: Saving rates: US (personal, *net*) vs euro area (household, *gross*) — compare the dynamics, not the levels.

area, while the marketable tier (T3, equities & funds) is $\sim$46% vs $\sim$19% — the US buffer is more sellable-fast but far more exposed to market price risk.

17 What this says about the question

1. **The false binary dissolves.** The post-2022 saving is best read as a *liquid precautionary buffer that also earns yield*. The "precaution versus yield-chasing" dichotomy was an artefact of treating only cash as liquid; on the ladder the broad sellable-fast share of saving flows *rose* (54% $\rightarrow$ 73%), and about half of household wealth is sellable within days (§11, §12).
2. **The operative uncertainty is income fear, not geopolitics.** Households' forward-looking expectations predict the saving rate ($R^2 \approx 0.38$ one quarter ahead) while the geopolitical index does not ($R^2 \approx 0$) — so the precautionary channel runs through perceived job/income risk (§14). Risk premia confirm that tension is priced (§13), but that is an enabling fact, not the household motive.
3. **The buffer is unevenly held.** The aggregate masks that the most energy-exposed, most income-fearful households have no cushion at all (§15).

The verdict therefore moves from the presented version's cautious "probably yield-chasing" to a sharper and better-defended statement: *a liquid precautionary buffer, optimised for yield once rates turned positive, driven more by income fear than by geopolitics, and concentrated among the households who can afford it.*

18 Limitations

- **Correlation, not causation.** Rates, inflation, and uncertainty all moved together after 2022; none of these descriptive results isolates the precautionary channel, and the main project's formal cross-country/IRF tests still return nulls.
- **Rate-consistency.** The money-supply term shift (§12) is equally consistent with a pure interest-rate story.
- **Mechanical component.** Part of the forward-looking R^2 reflects that the composite includes intended saving (§14).
- **Lower-bound liquidity.** Where Eurostat does not split F5, the equity/funds block is treated as illiquid, so the broad-liquid shares are lower bounds (§10).
- **Illustrative energy shares.** The by-income energy budget shares are stylised and should be verified against the Household Budget Survey.
- **Pending data.** Everything renders from free, non-FRED sources except two lines: the US *saving-rate* line and the credit/VIX series in the risk-premia figure, which fill in with FRED (the US ladder uses OECD; the sovereign spread uses the ECB fallback). All series are pulled live and revised — regenerate before citing.

Part III

Descriptive extension: the European saving picture

19 Why descriptive, and what these pictures show

The companion `../extension_feedback/` folder asks *why* the saving rate is high (a precautionary, liquidity-tiered reading). The supervisors asked to step back first and describe the phenomenon. This note does that with six self-contained charts, each a standalone script reusing the project helpers and the house style (shaded COVID / energy-hiking bands, a dashed Feb-2022 line, direct annotations, an italic footnote). It motivates, rather than tests, the hypothesis work.

20 Structural or cyclical?

Is the elevated rate a new *norm* or a *cycle* that should revert? A Hodrick–Prescott filter ($\lambda = 1600$) splits the saving rate into a slow trend (structural) and a cycle (Figure 12). Because the 2020–21 COVID spike was *forced* saving — households simply could not spend, not a behavioural change — those quarters are **excluded from the filter** (interpolated) so they do not distort the trend; the cycle is then the real deviation from that de-COVID trend, so the spike shows up (correctly) as a large positive *cycle*. On that basis the structural trend **steps up about** 2.6 **pp**, from 12.5% (2012–19) to 15.1% today, while the latest cycle is essentially *zero*. So the persistence is now **mostly structural**: a higher norm, consistent with saving more for old age as pension promises weaken, rather than a transient bump.

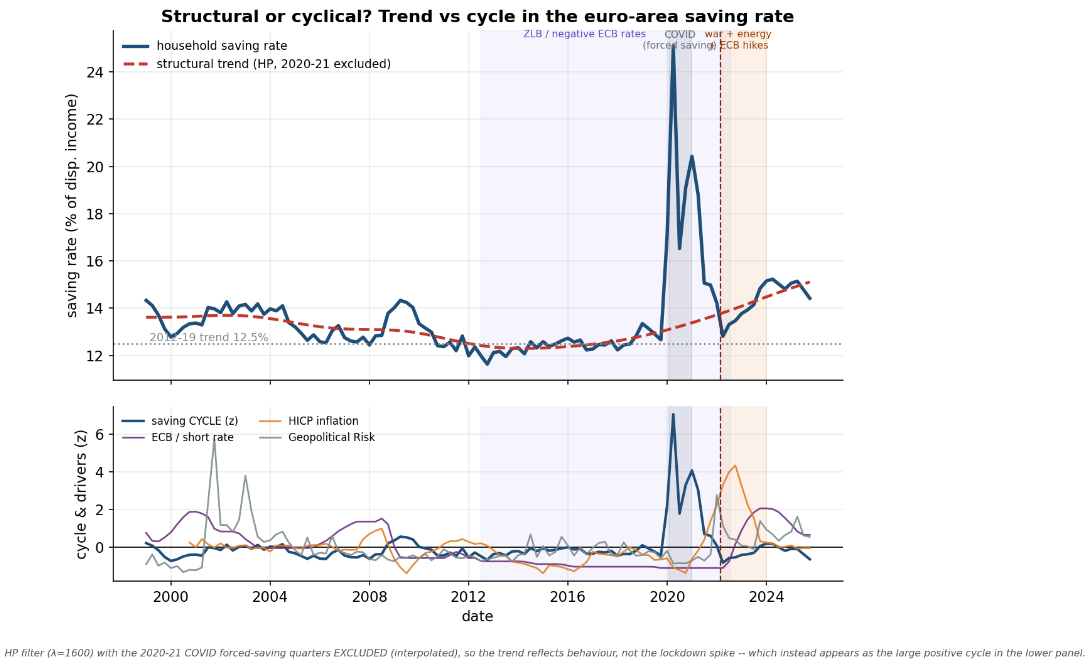


Figure 12: The euro-area saving rate split into a structural trend (HP, $\lambda = 1600$, with the 2020–21 COVID quarters excluded from the filter) and a cyclical component; the lower panel shows the cycle against cyclical drivers (ECB rate, inflation, GPR), z-scored.

21 The European map

Where is saving high? A choropleth of the gross household saving rate by country (Figure 13): **Germany** ∼20% **and France** ∼18%, the Nordics high; **Italy** ∼11%**, Spain** ∼13%, and **Greece dissaving**. The North–South gap is *structural and long-standing* — it predates the energy shock — so a raw "who saves most" map mostly reflects habits, not the recent crisis.

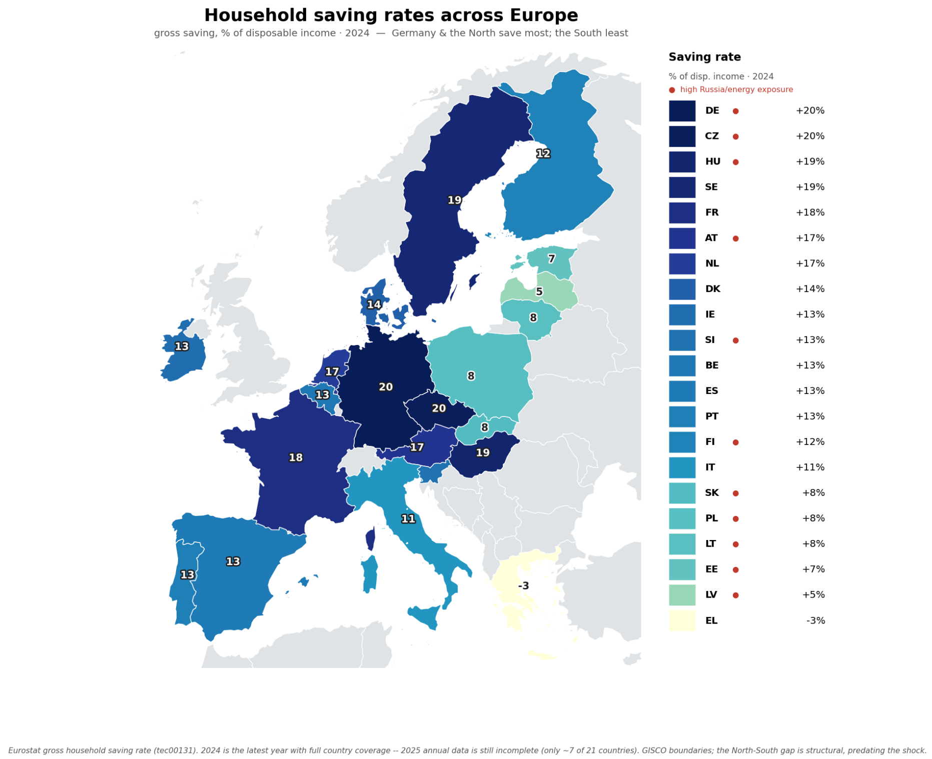


Figure 13: Gross household saving rate by country, 2024 — the latest year with full coverage (2025 is still incomplete); Eurostat TEC00131 with GISCO boundaries. North dark, South light; a ranked list gives every value.

The time dimension confirms the pattern (Figure 14): the country ranking is sticky — the North has saved roughly 8–12 pp more than the South for two decades — and the post-2022 lift raised most countries together rather than re-ordering them.

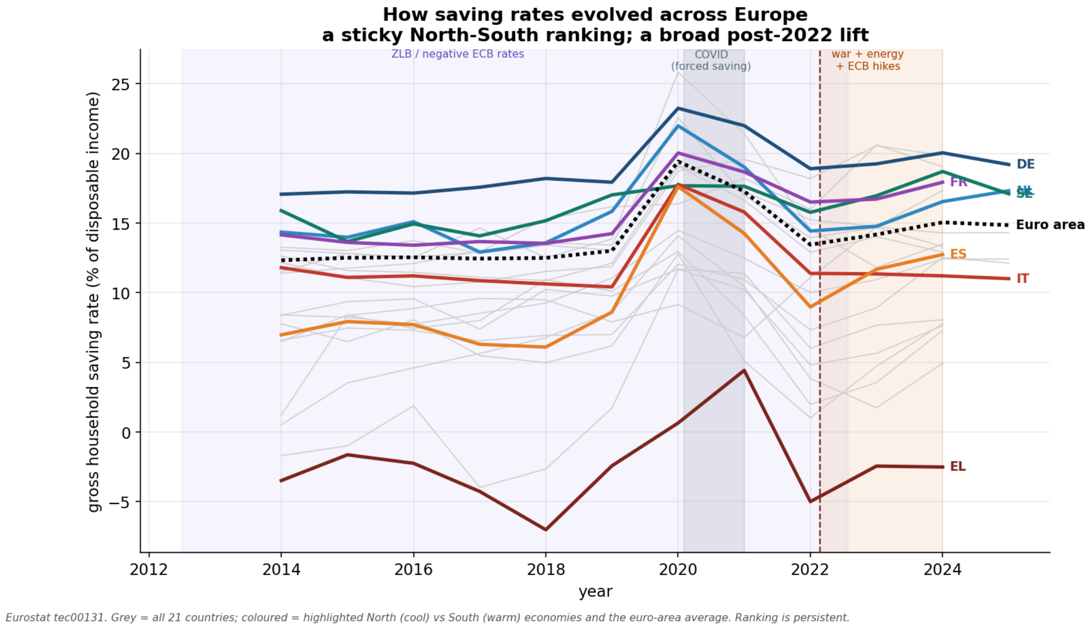


Figure 14: Household saving rates over time: all 21 countries (grey), with highlighted North (cool) and South (warm) economies and the euro-area average (black dotted); Eurostat TEC00131.

22 Spending: goods vs services

If saving is so high, how has services spending stayed so strong? It is not a paradox once goods and services are split (Figure 15). A high saving rate just means households spend a smaller *share* of income; *within* that spending the mix rotated. In the 2020 lockdowns **goods** boomed (durables, electronics) while **services** collapsed; after reopening **services rebounded** strongly (to ∼107, 2019=100) while **goods demand faded** (∼103). So "elevated services spending" is the post-COVID services recovery (plus high services inflation), not a consumer boom — and the goods weakness is part of why *overall* consumption, and hence the saving rate, stayed high. Strong services and high saving are reconciled by **weak goods**, not in tension.

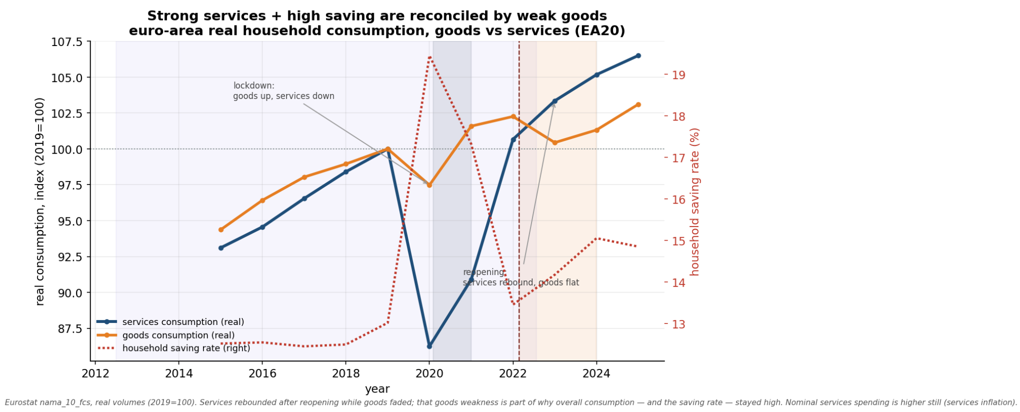


Figure 15: Real household consumption of goods vs services (2019=100, recent years) against the saving rate; Eurostat NAMA_10_FCS.

23 What households save in, and how it evolved

This is the *same* household balance sheet as the liquidity ladder, cut by *risk* rather than liquidity (Figure 16). The **risky share (equity & investment-fund shares, F5) rose about** $+8$ **pp over the decade,** 29% $\rightarrow$ 37%, while **non-risky deposits (F2) held near** 31%. Equity & funds **overtook deposits around 2021** and are now the largest single instrument, though deposits and insurance/pension claims (non-risky) still dominate combined. So the non-risky base is still large, but **equity involvement has clearly grown** — more euro-area households now carry market exposure. The cross-country snapshot (Figure 17) shows the risky share is higher in the core/North and lower in the South.

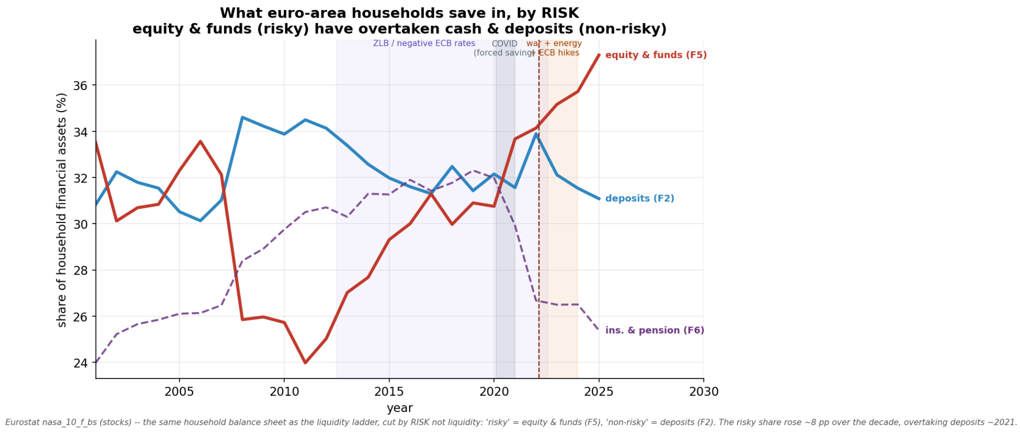


Figure 16: Household financial assets by RISK over time: non-risky deposits (F2), risky equity & funds (F5), and insurance & pensions; Eurostat NASA_10_F_BS. Risky overtook non-risky around 2021.

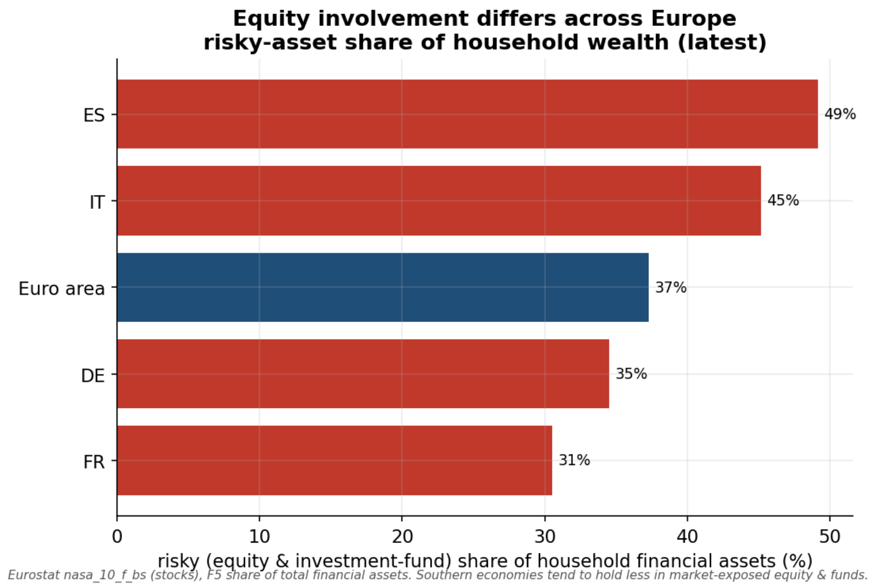


Figure 17: Risky (equity & investment-fund) share of household financial assets, by country (latest).

24 Demographics and pensions

Do aging and pension design move household saving? Two cross-country scatters (Figure 18). *(a)* The old-age dependency ratio is, if anything, *weakly negatively* related to the saving rate ($r \approx -0.14$): older Southern economies save *less*, not more, so aging does not mechanically lift saving. *(b)* A funded-pension proxy — the insurance & pension share of household assets (F6), high where households actually *hold* pension assets and low under pay-as-you-go state promises — is *weakly positively* related to saving ($r \approx +0.34$): big funded systems (Netherlands, Denmark) coexist with solid saving, not low saving. So "good pensions ⇒ less private saving" is not clear in the cross-section — pensions shape *where* wealth sits (funded systems hold more in F6/F5) more than *how much* is saved.

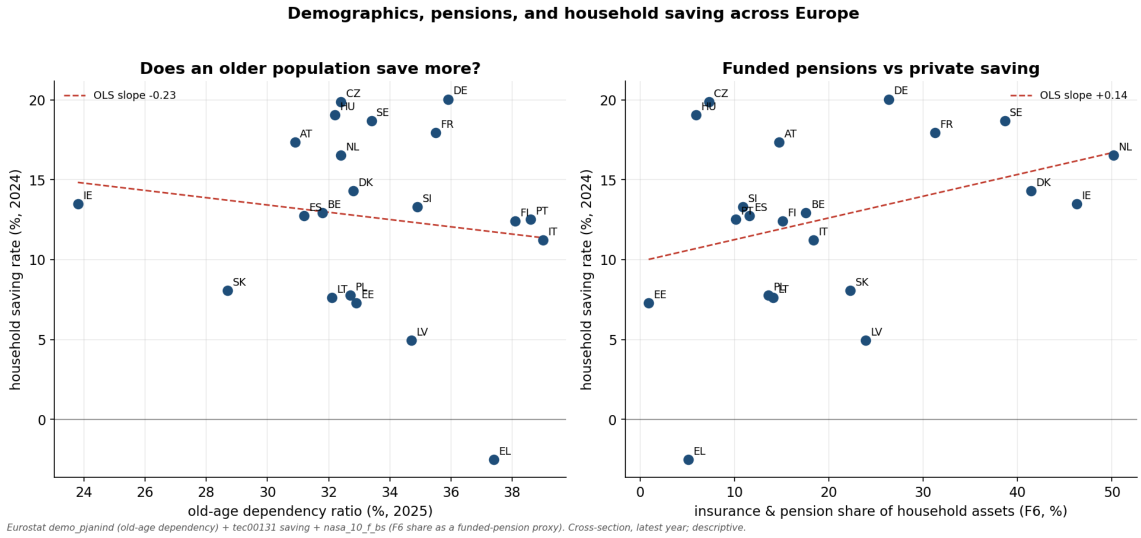


Figure 18: Cross-country: old-age dependency vs the saving rate (left) and the funded-pension proxy (F6 share) vs the saving rate (right), latest year; Eurostat DEMO_PJANIND, TEC00131, NASA_10_F_BS.

25 Why Europeans look like cautious savers

The clearest US–Europe difference is not how *much* households save but *how* they hold it (Figure 19). US households hold about **57%** of their financial wealth in equity & investment funds versus ~37% in the euro area — roughly 1.5× — with the mirror image in deposits (US ~11% vs euro area ~31%). The gap is large and *persistent* across the last two decades. The institutional history behind it: the US built a funded, equity-heavy private-pension system (401(k)/IRA) and an equity-investing culture, while much of continental Europe leaned on *pay-as-you-go state pensions* and a bank-deposit culture. European households therefore carry less market risk and look more "cautious" — a structural, pension-rooted difference, not merely preferences.

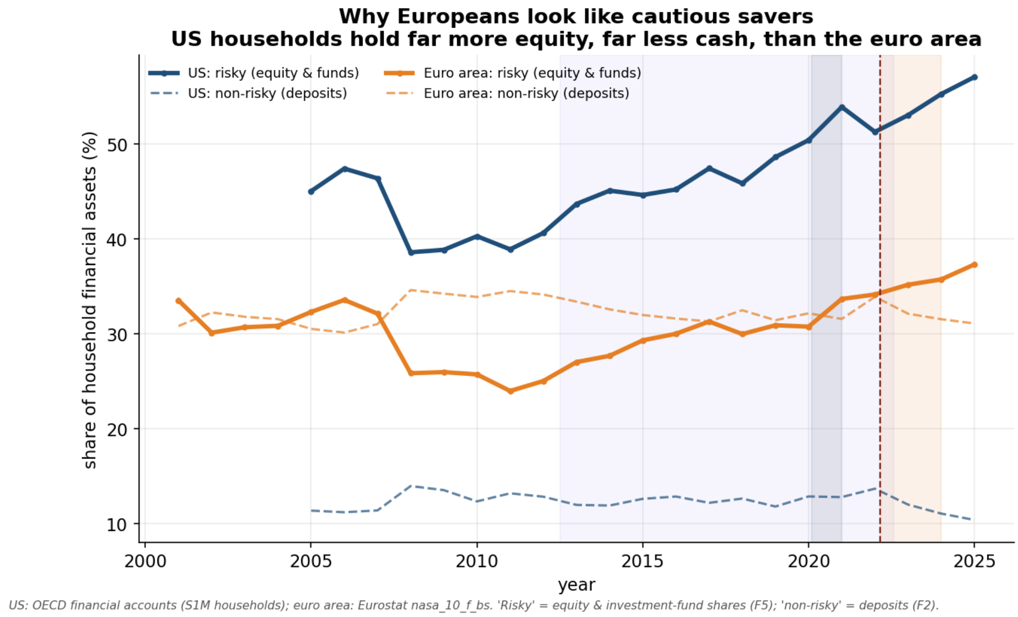


Figure 19: Risky (equity & funds, F5) and non-risky (deposits, F2) shares of household financial assets, US vs euro area, over time. US: OECD financial accounts; euro area: Eurostat NASA_10_F_BS.

26 Synthesis

1. The elevated saving rate is now a **higher structural norm**, not a passing cycle (§20).
2. It sits on a **structural North–South map** that long predates the shock (§21).
3. High saving coexists with **strong services** via a goods→services rotation (§22).
4. Households still hold a large **non-risky** base, but **equity involvement has grown** markedly over the decade (§23).
5. **Aging and pension design** move household saving only weakly in the cross-section; pensions shape *where* wealth sits more than how much is saved (§24).
6. Versus the US, euro-area households are not low savers so much as **cautious holders** — far less equity, far more cash — a structural, pension-rooted difference (§25).

These descriptive facts motivate the hypothesis work in `../extension_feedback/`: a precautionary, liquidity-tiered reading of why the rate is high and where the money sits. All series are pulled live and revised — regenerate before citing.

A Reproduction details

A.1 File tree

```
figures/                # ALL charts, one folder, tagged by idea letter A-O
data/                   # ALL generated data, prefixed with the matching idea letter
code/                   # analysis scripts, grouped by module:
  core/                 #   A-D  saving vs uncertainty, country, distribution
    (01/02/03)
  robustness/           #   E    panel FE, event study, local projections
  follow_money/         #   F    where the money went (composition tests)
  feedback/             #   G-N  liquidity ladder, money supply, risk premia,
                        #        forward-looking, energy, US, reconciliation,
    consumption
  descriptive/          #   O    structural/cyclical, maps, demographics
  legacy/               #        superseded original scripts (safe to delete)
documentation.tex       # this document (core + feedback + descriptive merged)
findings.md             # narrative write-up of every idea
series_catalog.md       # verified macro time-series catalogue (liquidity ladder)
article-ideas.md        # blog-angle working notes
run_all.sh, requirements.txt   # core pipeline + pinned deps
NOTE: figures/data now carry an idea-letter prefix (A_, B_, ... O_); scripts moved
under code/<module>/, so in-script paths may need adjusting before a fresh run.
```

A.2 Commands

```
# one-shot, fully reproducible
bash run_all.sh

# or stage by stage (after: pip install -r requirements.txt)
python 01_collect_data.py
python 02_make_figures.py
python 03_econometrics.py            # GPR (default)
python 03_econometrics.py --proxy epu --substart 2010
```

A.3 Data dictionary (generated CSVs)

file	frequency	contents
`ea_saving_rate_quarterly.csv`	quarterly	date, saving (% GDI)
`gpr.csv`	monthly	date, GPR index
`fred_eu_epu.csv`	monthly	date, EU EPU index
`ecb_rate.csv`	daily	date, ECB deposit rate (%)
`ea_inflation.csv`	monthly	date, HICP YoY (%)
`country_saving_annual.csv`	annual	geo $\times$ year saving pivot
`country_energy_peak_2022.csv`	2022 scalar	geo, peak energy infl. (%)
`scatter_energy_vs_saving.csv`	cross-section	geo, saving rise, energy peak
`saving_rate_by_quintile.csv`	snapshot	q, value, geo, year
`icw_quintile_panel.csv`	snapshots	year, q, value, geo
`ces_unemp_expectations.csv`	latest	group, expected unemployment (%)
`econometrics_results.md`	—	full text log of step 3

All numerical results in this document come from a clean end-to-end run and are reproducible via `run_all.sh`. Because the underlying series are revised by their providers, regenerate the data and re-compile before citing.